

Abstract

This project presents the development of a console-based Tic Tac Toe game using C++. Tic Tac Toe is a classic two-player game played on a 3×3 grid, where players alternately place their respective symbols, 'X' and 'O', with the objective of forming a continuous line either horizontally, vertically, or diagonally.

The application is implemented using fundamental C++ programming constructs, including arrays, control structures, and functions. A two-dimensional array is used to represent the game board, while conditional logic is applied to validate moves and determine the game outcome after each turn. The program ensures user interaction through input prompts and displays the updated game board dynamically.

The primary objective of this project is to reinforce core programming concepts such as logical reasoning, modular design, and user input handling. It also introduces students to basic game development techniques within a structured programming environment. Overall, the project serves as an effective educational tool for beginners to understand and apply essential C++ principles in a practical context.